PROJECT PROPOSAL

InsightFlow AI – Business Meeting Intelligence Platform

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OVERVIEW

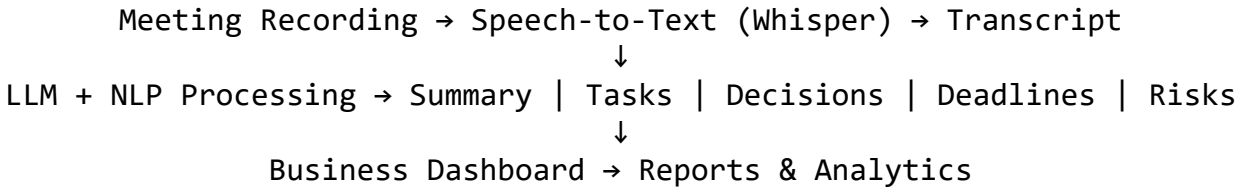
Business meetings generate valuable discussions, decisions, and action items, yet most organizations still rely on manual note-taking, which is time-consuming, inconsistent, and prone to lost information. InsightFlow AI is proposed as an AI-powered platform that will convert meeting recordings into structured, actionable business intelligence, automatically generating transcripts, summaries, action items, decisions, deadlines, and risk flags, and presenting them through an interactive dashboard. The platform is intended for project managers, team leads, and organizational leadership who require a reliable, centralized record of meeting outcomes to improve accountability and decision-making.

PROBLEM STATEMENT & OBJECTIVES

Manual documentation leads to incomplete records, unclear task ownership, missed deadlines, and no centralized way to revisit past decisions. Existing meeting tools largely stop at recording or basic transcription, leaving a gap between discussion and execution. The project therefore aims to automate meeting documentation, improve decision and task tracking, enhance accountability through clear owner–deadline mapping, reduce manual administrative effort, and support data-driven decision-making through AI-generated insights and analytics.

PROPOSED SOLUTION & WORKFLOW

The platform will accept an uploaded meeting recording and convert it to text using OpenAI Whisper. The transcript will then be processed with Large Language Models and NLP techniques to extract summaries, action items, owners, deadlines, decisions, and risks, which will be stored and visualized through a web-based dashboard, as outlined below.



EXECUTION PLAN & METHODOLOGY

The project will follow a modular pipeline architecture, developed and validated in stages:

(1) user authentication and secure meeting upload

(2) speech-to-text transcription via Whisper

(3) AI-based summarization and information extraction using LLMs, spaCy, and Transformers

(4) storage of structured results in a PostgreSQL database
(5) visualization through a React-based dashboard built with Plotly, exposed via FastAPI backend services.
Each module will be built and tested independently before integration into the unified prototype, using Git/GitHub for version control and Docker for consistent deployment. Development, evaluation, and testing will use sample and self-recorded meeting datasets, with organizational data used only where permitted.

TECHNOLOGY STACK

Technology	Role in the Project
Python / FastAPI	Backend logic and APIs
React.js / Plotly	Interactive dashboard and analytics
PostgreSQL	Structured data storage
OpenAI Whisper	Speech-to-text transcription
LLMs / spaCy / Transformers	Summarization, NLP, and information extraction
Docker / GitHub	Deployment and version control

EXPECTED OUTCOME & CONCLUSION

The prototype is expected to demonstrate faster, more consistent meeting documentation, improved follow-up on action items, stronger accountability, and centralized, analyzable meeting records. By combining speech recognition, NLP, and LLMs into a single workflow, InsightFlow AI aims to show a practical, scalable application of AI to a common organizational challenge, laying the foundation for future enhancements such as calendar integration, multilingual support, and conversational querying across past meetings.

Project Mentor
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